Tokenization Efficiency in Ghanaian Low-Resource Languages: A Comparative Evaluation of Modern Multilingual Tokenizers for Twi and Ewe

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Repository: `https://github.com/exornamb/GhanaToken` (final tag: `v1.0-final`)

Abstract. Language models cut text into small pieces called tokens. When a tokenizer was built mostly from large languages, words in smaller languages are split into many pieces, raising cost and shrinking the amount of text a model can read at once. This study measures the problem for Twi and Ewe, two widely spoken Ghanaian languages, using parallel text from the GhanaNLP corpora (6,072 Twi and 3,456 Ewe cleaned sentences). Four existing tokenizers (mBERT, XLM-R, AfroXLMR and the Twi-specific ABENA) split 64% to 72% of words into several pieces on sealed test sentences and use 2.29 to 2.61 tokens per word. GhanaTok, a small 8,000-token tokenizer shared by both languages, was then trained, keeping Twi and Ewe letters inside words and reserving space for common word parts and greetings. GhanaTok uses 30.7% fewer tokens per word for Twi and 23.6% fewer for Ewe than the best general tokenizer (mBERT), and 31.5% fewer than ABENA on Twi. All five main comparisons stay significant after correction for multiple tests. A Unigram version cuts 41% to 49%. Sentence-level error analysis shows GhanaTok using fewer tokens than mBERT on 98.4% of Twi test sentences, with the remaining failures concentrated in English phrases, loanwords and quotation marks. Gains flatten after 8,000 tokens on this corpus, and a small sentiment test shows no clear accuracy difference between tokenizers. Better language model quality is not shown and remains untested.

Keywords: Tokenization · Low-resource languages · Twi · Ewe · Error analysis

1 Introduction

Language models do not read words directly. Text is first cut into small pieces called tokens, using a fixed list of pieces called a vocabulary. When the list is built mostly from large languages such as English, words from smaller languages are often cut into many short pieces. The average number of tokens per word is called *token fertility*. High fertility has real costs. Services charging by the token become more expensive and can read less text for these languages [2, 20], and higher fertility goes together with lower accuracy across 16 African languages [14]. Tokenizers are also known to treat languages unequally [17].

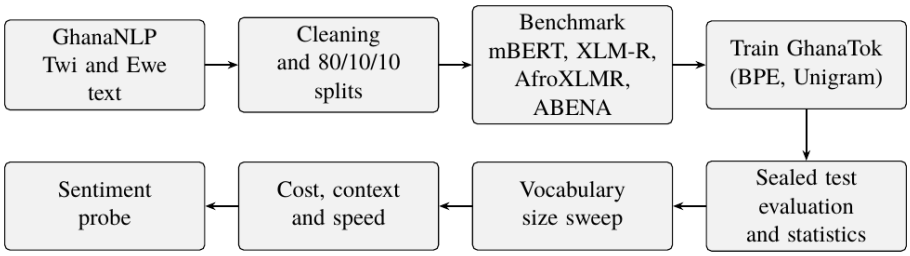


Fig. 1: An overview of the study. Existing tokenizers are benchmarked, GhanaTok is trained on the training splits only, and all tokenizers are compared on sealed test sentences, followed by error analysis and three extensions.

Twi (also called Akan) and Ewe are widely spoken in Ghana. Both use letters absent from standard English text, such as ɛ, ɔ, ɖ, ŋ, ʋ and ƒ. Twi already has language models such as ABENA [5], and the GhanaNLP corpora now give parallel text for five Ghanaian languages [1]. A survey of the field still describes Ghanaian languages as under-served [11]. A controlled, side-by-side measurement of common tokenizers on the same Twi and Ewe text is missing, as is a test of whether a small tokenizer built for just these two languages fixes the problem and where such a tokenizer still fails.

Research questions. Three questions guide the work. **RQ1:** How badly do general multilingual tokenizers (mBERT, XLM-R, AfroXLMR) and a regional one (ABENA) fragment Twi and Ewe text? **RQ2:** Does a small dedicated tokenizer, GhanaTok, cut fertility and fragmentation on sealed test text? **RQ3:** Can one shared 8,000-token vocabulary work well for both languages? A fourth question, whether lower fertility improves downstream accuracy, was deliberately left to future work; only a small supplementary probe touches the topic.

Present work. This paper measures how four existing tokenizers split Twi and Ewe text, tests GhanaTok, and analyses where GhanaTok succeeds and fails. The key contributions can be summarized as follows:

- A same-data benchmark of four existing tokenizers on Twi and Ewe, using fertility, tokens per character, and the share of words that get split.
- GhanaTok, a shared 8,000-token tokenizer in a byte-level BPE version and a SentencePiece Unigram version, tested on sealed sentences with paired significance tests and confidence intervals.
- A sentence-level error analysis, a vocabulary size sweep, cost and context estimates, and a small sentiment check, showing how far the gains go and where the gains stop.

The study does not claim that tokenizer unfairness is a new finding. The aim is a clear, reproducible Ghanaian case study.

The rest of the paper is structured as follows. Section 2 reviews related work, while Section 3 describes the data and ethics. Section 4 presents the methods,

including the baselines, GhanaTok, the measures and the statistics. The results are reported in Section 5 and examined through error analysis in Section 6. Section 7 discusses the findings, Section 8 lists limitations, and Section 9 concludes. Code and data availability, the AI-use statement and the references close the paper.

2 Related Work

2.1 Tokenizer Unfairness and the Cost of Fertility

Petrov et al. [17] show that the same text needs very different numbers of tokens in different languages, and Ahia et al. [2] link the gap to what users pay for commercial models. Lundin et al. [14] find that fertility predicts accuracy across 16 African languages and that doubling the tokens roughly quadruples training cost. Somide [20] measures 20 African languages on 11 tokenizers: every language costs more than English, with a median premium of 1.88 times on GPT-5's tokenizer and up to 8.9 times the inference cost, and the paper concludes that tokenizers designed with African languages in mind are needed to bring the gap down. The cost estimates in Section 5.5 follow the same idea for two Ghanaian languages.

2.2 Tokenizer Design

Subword methods such as BPE [19] and Unigram [12], packaged in SentencePiece [13], are the standard tools. Rust et al. [18] show that tokenizer quality affects monolingual performance, and Kautsar and Koto [3] train one tokenizer per language and align the vocabularies with dictionaries, improving fertility balance across 13 low-resource languages. This study takes the simplest route: one small vocabulary trained only on the target languages.

2.3 Models and Resources for Ghanaian Languages

mBERT [7], XLM-R [6] and AfroXLMR [4] are multilingual models whose tokenizers serve as general baselines. Issaka et al. [11] survey NLP for Ghanaian languages, and Issaka et al. [10] describe a large collaborative data effort for African languages that includes Twi. ABENA [5] is the best known Twi model, and AfriSenti [16] offers Twi sentiment data. No study found in this review compares general and regional tokenizers on the same Twi and Ewe text, tests significance with paired statistics, and inspects the remaining failures of a small custom vocabulary. This project addresses that gap.

Table 1: Number of sentences per language after cleaning.

Language	Raw	Cleaned	Train	Validation	Test
Twi	6,090	6,072	4,857	607	608
Ewe	3,488	3,456	2,764	345	347

3 Data and Ethics

3.1 Parallel Text

The Twi and Ewe sides of the GhanaNLP parallel corpora [1] are used. Cleaning applies Unicode NFC normalisation, whitespace cleanup and duplicate removal, and keeps sentences with at least two words. Each language is split 80/10/10 into train, validation and test sets (Table 1). The test sets stay sealed until the final run. The first comparison of existing tokenizers uses a calibration sample of 3,000 sentences per language from the train and validation splits. The text leans toward everyday dialogue, Bible text and civic notices, and the Twi side mixes Akuapem and Asante spelling.

3.2 Sentiment Data

The supplementary probe uses the Twi part of AfriSenti [16]: 4,818 tweets with three labels (positive 2,277, negative 1,815, neutral 726) and official splits of 3,481 for training, 388 for validation and 949 for testing.

3.3 Ethics and Responsible Use

All text is public and was created for research: the GhanaNLP parallel corpora (CC BY-NC-SA 4.0) and the AfriSenti tweets (CC BY 4.0), used here for academic research only. No new text was collected, no writer was identified, and results are reported as averages over many sentences. Example sentences in this paper are taken from the released corpus. Tokenizer files trained on the GhanaNLP text should be shared under compatible terms. Claims cover Twi and Ewe only as represented in this data, with no statement about how all Ghanaians speak or write. The main risks are over-claiming (lower token counts read as better model quality), an unfair comparison (GhanaTok was trained on the same corpus as the test text), cost estimates read as real bills, and the absence of review by native speakers. Each risk is addressed in the text and listed in Section 8.

3.4 Data Access and Restrictions

The repository contains code, notebooks, saved tables, figures and documentation. Raw and processed text files are not committed, in line with the redistribution terms of the sources. The data are public and obtained from the Hugging Face Ghana-NLP organisation (GhanaNLP corpora) and the

`masakhane/afrisenti` dataset, following the notebooks and the README. No private or restricted data, passwords or API keys are used or stored.

4 Methods

4.1 Baselines

mBERT uses WordPiece. XLM-R uses a SentencePiece vocabulary of 250,000 tokens. AfroXLMR is XLM-R adapted to African languages. ABENA was trained on Akuapem Twi only, so the tokenizer is tested on Twi and never on Ewe. XLM-R and AfroXLMR gave identical token counts on every test sentence. These four tokenizers form the comparison for RQ1 and RQ2.

4.2 GhanaTok

GhanaTok is built from four design choices.

Vocabulary. GhanaTok has 8,000 tokens shared by Twi and Ewe. The tokenizer is trained on the combined training splits (7,621 sentences, about 649,000 characters). A small vocabulary was picked because the corpus is small, and Section 5.4 tests that choice.

Letter-aware pre-tokenizer. A custom rule keeps the extended letters ɛ, ɔ, ɖ, ŋ, ʋ, ƒ and tone marks attached to the word, so the letters are not treated as places to split.

Seed words. Vocabulary slots are reserved for common prefixes and suffixes (for example *mbe-*, *bɛ-*, *-foɔ*, *-wo*, *-m*) and greetings (*Akwaaba*, *Medaase*, *Woezɔ*, *Aleke*), each in lower case and capitalised form. Two bugs were fixed during development: seed words were first missing in capitalised form, and decoding with special tokens skipped dropped seed words.

Two versions. **GhanaTok (BPE)** is the main model, trained with byte-level BPE and a minimum pair frequency of 2. **GhanaTok-Unigram** uses SentencePiece Unigram with full character coverage, on the same data and seed words. The Unigram version serves as a secondary comparison.

4.3 Measures

For a sentence s with t_s tokens (special tokens excluded) and w_s whitespace-separated words, *token fertility* over S sentences is

$$F = \frac{1}{S} \sum_{s=1}^{S} \frac{t_s}{w_s}. \tag{1}$$

Tokens per character and the *fragmentation rate* (the percentage of words split into two or more tokens) are also reported. Lower is better for all three. Pooling all words instead of averaging per sentence gives slightly lower values for every

tokenizer and the same ranking: for Twi, GhanaTok gets 1.647 and mBERT 2.354 (a 30.0% cut); for Ewe, 1.745 and 2.245 (22.3%).

4.4 Statistical Analysis

All tokenizers see the same sentences in the same order, so paired Wilcoxon signed-rank tests [21] and paired bootstrap 95% confidence intervals with 1,000 resamples [8] are used. The five main RQ2 tests are corrected with the Holm method [9]. Twi and Ewe test sets are different sentences, so languages are compared with a Mann-Whitney U test [15] and an unpaired bootstrap interval. Bootstrap intervals go beyond the point estimates common in earlier tokenizer studies and suit the modest size of the test sets.

4.5 Extensions

Vocabulary size is varied at 4,000, 8,000, 16,000 and 32,000 tokens for both versions. Cost is estimated from fertility with a reference price of USD 0.15 per million input tokens, and context capacity is the number of words fitting in windows of 4,000, 32,000 and 128,000 tokens. Speed is the time to encode a whole test set over 5 repeated runs after 1 warm-up run. The sentiment probe trains a bag-of-tokens logistic regression (random state 42, at most 1,000 iterations) for each tokenizer on the AfriSenti training and validation tweets and reports accuracy, macro F1 with a bootstrap interval, and the number of features used.

5 Results

5.1 Existing Tokenizers (RQ1)

On the calibration sample, mBERT gives the lowest fertility (Twi 2.371, Ewe 2.303). ABENA gets 2.402 on Twi, and XLM-R and AfroXLMR get 2.438 on Twi and 2.625 on Ewe. Between 64.7% and 71.7% of words are split. Table 2 shows the same pattern on the sealed test sentences: 2.29 to 2.61 tokens per word, and 64% to 72% of words split. ABENA, built for Twi, does no better on Twi than the best general tokenizer (2.419 against 2.389 for mBERT).

5.2 GhanaTok (RQ2)

GhanaTok (BPE) reaches 1.656 tokens per word on Twi and 1.750 on Ewe, and splits about half of all words (48.8% and 51.3%). Table 3 shows the cuts against each baseline with the intervals. Against the best general tokenizer, mBERT, the cut is 30.7% on Twi and 23.6% on Ewe. Against ABENA on Twi the cut is 31.5%. All five main tests stay significant after Holm correction. GhanaTok-Unigram goes further: 1.251 on Twi and 1.349 on Ewe, 47.6% and 41.1% below mBERT and about 48% below XLM-R in both languages. On Twi words seen in the test set (939 unique), Unigram gives a shorter split on 597, BPE on 50, and 292 are tied.

Table 2: Sealed test results: fertility (tokens per word) and percentage of words split. Lower is better.

	Twi (608)		Ewe (347)	
Tokenizer	Fert.	% split	Fert.	% split
mBERT	2.389	70.5	2.291	63.9
XLM-R	2.444	70.3	2.608	71.0
AfroXLMR	2.444	70.3	2.608	71.0
ABENA	2.419	72.1	n/a	n/a
GhanaTok (BPE)	1.656	48.8	1.750	51.3
GhanaTok-Unigram	1.251	17.6	1.349	22.8

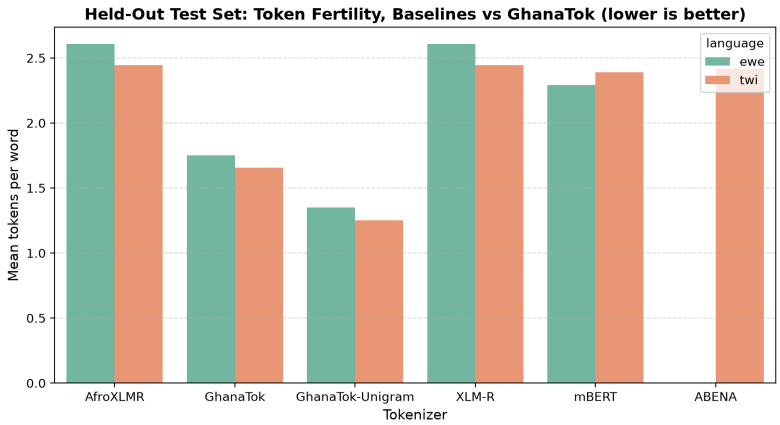


Fig. 2: Token fertility on the sealed test sets.

5.3 One Vocabulary for Both Languages (RQ3)

GhanaTok scores 1.656 on Twi and 1.750 on Ewe. The gap of 0.094 tokens per word (95% interval 0.061 to 0.128) is statistically significant (Mann-Whitney $U = 79{,}394.5$, $p = 1.9 \times 10^{-10}$) but small. Unigram shows the same pattern (gap 0.098). Percent cuts are larger on Twi mostly because mBERT is already relatively good on Ewe (2.291 against 2.389 on Twi). Because the two test sets are different sentences, the gap may also reflect content and not only language.

5.4 Vocabulary Size

Both versions were trained at 4,000, 8,000, 16,000 and 32,000 tokens and each was tested on the sealed sets (Fig. 3). Going from 4,000 to 8,000 tokens cuts fertility by 5.2% to 9.7%. Going from 8,000 to 16,000 cuts fertility by only 2.2% to 4.3%, and 32,000 gives exactly the same result as 16,000. The training logs show that this corpus holds only 10,937 distinct pieces for BPE and 11,639 for Unigram, so a larger target cannot be filled. At 8,000 tokens, fertility is within

Table 3: Fertility cut by GhanaTok (BPE) on the sealed test sets, with paired bootstrap intervals and Holm-corrected Wilcoxon p values. Unigram rows are in Table 9.

Lang.	Baseline	Base	GhanaTok	Cut % [95% CI]	Holm p
Twi	mBERT	2.389	1.656	30.7 [29.85, 31.44]	$< 10^{-50}$
Twi	XLM-R / AfroXLMR	2.444	1.656	32.2 [31.45, 32.98]	$< 10^{-50}$
Twi	ABENA	2.419	1.656	31.5 [30.72, 32.30]	$< 10^{-50}$
Ewe	mBERT	2.291	1.750	23.6 [22.08, 24.95]	$< 10^{-50}$
Ewe	XLM-R / AfroXLMR	2.608	1.750	32.9 [31.61, 34.11]	$< 10^{-50}$

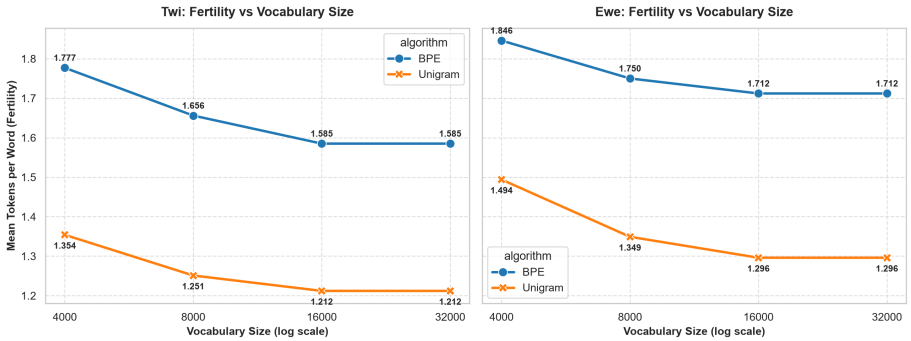


Fig. 3: Fertility against vocabulary size on the sealed test sets.

2.2% to 4.5% of the best value seen in every case. The sweep was run on the sealed test sets after 8,000 had already been chosen, so the sweep supports the choice but is not an independent selection.

5.5 Cost and Context

Token counts decide what many services charge and how much text fits in a context window. Table 4 turns the fertility numbers into price-free multipliers. For the same Twi text, mBERT, ABENA and XLM-R would produce 44% to 48% more tokens than GhanaTok (BPE), and for Ewe the extra is 31% (mBERT) to 49% (XLM-R). A 4,000-token window would hold 778 more Twi words and 752 more Ewe words with GhanaTok (BPE) than with XLM-R. At a reference rate of \$0.15 per million input tokens, one million Twi words would cost about \$0.25 with GhanaTok (BPE) and \$0.37 with XLM-R. These figures are estimates of token counts. Commercial services use separate tokenizers, so the figures are not measured bills.

5.6 Speed and Sentiment Probe

Whole-test-set encoding times (Table 12) vary a lot between tokenizers and runs, and GhanaTok (BPE) has a large spread on Twi. The times come from one computer and different software libraries, and the causes were not separated, so

Table 4: Cost relative to GhanaTok (BPE) and words fitting in a 4,000-token window.

Lang.	Tokenizer	Fert.	Cost	Words in 4k
Twi	GhanaTok-Unigram	1.251	0.76x	3,197
Twi	GhanaTok (BPE)	1.656	1.00x	2,415
Twi	mBERT	2.389	1.44x	1,674
Twi	ABENA	2.419	1.46x	1,654
Twi	XLM-R / AfroXLMR	2.444	1.48x	1,637
Ewe	GhanaTok-Unigram	1.349	0.77x	2,965
Ewe	GhanaTok (BPE)	1.750	1.00x	2,286
Ewe	mBERT	2.291	1.31x	1,746
Ewe	XLM-R / AfroXLMR	2.608	1.49x	1,534

the times are treated as rough. On the AfriSenti Twi tweets, the probe gives 62.9% accuracy with either GhanaTok version and 64.1% to 65.0% with the others (Table 13). The macro F1 intervals overlap for all six tokenizers, so no tokenizer can be called better. GhanaTok uses about half as many active features (1,899 and 2,133 against 3,886 to 4,483). The mBERT and ABENA results are almost identical (macro F1 0.5624 for both, with per-class scores differing only in the fourth decimal), and the cause was not isolated. The probe is a rough check with a simple model on short tweets, not a test of language model quality.

6 Error Analysis

Averages hide where a tokenizer fails. The analysis joins the per-sentence results to the sealed Twi test sentences (word and character counts match for all 608 sentences) and compares GhanaTok (BPE) with each baseline sentence by sentence. The code is `src/error_analysis.py` and the outputs are saved in `reports/tables`. Ewe sentences are analysed by counts only, because the analysis used the Twi test text.

6.1 Sentence-Level Wins and Losses

GhanaTok uses fewer tokens per word than mBERT on 598 of 608 Twi sentences (98.4%), ties on 5 and loses on 5 (Table 5). Against XLM-R the figures are 603 wins, 1 tie and 4 losses, and against ABENA 601 wins, 3 ties and 4 losses. On Ewe, GhanaTok wins on 330 of 347 sentences against mBERT (95.1%) but loses on 13 (3.7%), a higher loss rate than on Twi. The average gain is therefore not driven by a few sentences: the gain holds on almost every sentence. Fertility for GhanaTok has a median of 1.625 on Twi and 1.714 on Ewe, a 90th percentile of 1.92 and 2.00, and a maximum of 3.0 in both languages. Only 38 Twi sentences (6.2%) and 30 Ewe sentences (8.6%) exceed 2 tokens per word.

Table 5: Sentence-by-sentence comparison of GhanaTok (BPE) with each baseline on the sealed test sets.

Language	Baseline	GhanaTok fewer	Tied	GhanaTok more
Twi	mBERT	598 (98.4%)	5 (0.8%)	5 (0.8%)
Twi	XLM-R	603 (99.2%)	1 (0.2%)	4 (0.7%)
Twi	ABENA	601 (98.8%)	3 (0.5%)	4 (0.7%)
Ewe	mBERT	330 (95.1%)	4 (1.2%)	13 (3.7%)
Ewe	XLM-R	344 (99.1%)	1 (0.3%)	2 (0.6%)

6.2 Hardest Sentences

Table 6 lists the six Twi sentences with the highest GhanaTok fertility. Three patterns appear.

English phrases and loanwords. Of the five Twi sentences where GhanaTok loses to mBERT, four contain English phrases (for example *Pan African Literacy For All Conference*, *Standard operation procedures*, *Ghana Virus Research Institute* and a full English clause) and the fifth contains the loanword *sigaret-nom*. The hardest sentence contains the written form *fisiks* with the English word *physics* in brackets. GhanaTok was trained only on Twi and Ewe text, so English words have few vocabulary entries, while mBERT's vocabulary already holds many English pieces. This explanation fits the pattern but was not tested directly.

Quotation marks and digits. Of the 608 Twi sentences, 94 contain quotation marks, and these have a mean fertility of 1.777 against 1.634 for the rest. Among the 61 sentences in the top fertility decile, 4.9% contain digits against 2.3% overall, and the mean number of capitalised words is 1.64 against 1.20. These are observations, not causal findings, because quotation marks, capitals and digits may travel with other features such as names and mixed-language text.

Sentence length. The rank correlation between sentence length and fertility is small and negative for Twi (-0.121, $p = 0.003$) and not significant for Ewe (-0.037, $p = 0.50$). Short sentences cost more for every tokenizer (Table 7), because a single capitalised greeting or name carries a larger share of the words, but the effect is modest.

6.3 Word-Level Examples and Dialect Check

Word-level examples (Table 8) show names such as *Nkrumah* and *Mprempren* kept whole by GhanaTok but cut into three pieces by XLM-R, and the Ewe word *Afetɔ*, containing ƒ, kept whole against four pieces for XLM-R. Some words stay fragmented: *Ahwɛyie* takes three tokens in both GhanaTok versions. For Akuapem spelling, 76 of 608 test sentences (12.5%) matched a small keyword list. ABENA's fertility on these sentences is 2.377 against 2.425 on the rest, a small gap in the expected direction, while GhanaTok shows almost no gap (1.663 against 1.655). The check uses keywords and is not a dialect classifier.

Table 6: The six Twi test sentences with the highest GhanaTok (BPE) fertility, compared with mBERT. Fertility is tokens per word.

Rank	Words	GhanaTok	mBERT	Sentence
1	9	3.00	3.56	Asukuufoɔ no bɛyɛ 'fisiks' (physics) adesua dwumadie ɔkyena anɔpa.
2	7	2.71	4.86	"Sini no kyerɛkyerɛɛ asomdwoeɛ, pɛpɛɛpɛyɛ ne nkɔsoɔ."
3	6	2.67	3.17	Mma nkurɔfo nsɛe w'asomdwoe beae no.
4	7	2.57	3.14	Mmoa aduane ɛwɔ ayuo nnuanenuru bebree wɔm.
5	7	2.57	2.43	Asanom ne sigaretnom nyɛ papa mma w'akwahosan.
6	25	2.52	1.72	"The chief administrative officer berieves there is a lot of money not acco...

Table 7: Fertility by sentence length in words. AfroXLMR matches XLM-R and is omitted.

Lang.	Tokenizer	1 to 5	6 to 10	11 to 20	21+
Twi	mBERT	2.700	2.554	2.339	2.312
Twi	XLM-R / AfroXLMR	2.700	2.586	2.409	2.340
Twi	ABENA	2.600	2.578	2.374	2.335
Twi	GhanaTok (BPE)	2.100	1.736	1.621	1.663
Twi	GhanaTok-Unigram	1.500	1.333	1.214	1.266
Ewe	mBERT	2.800	2.486	2.303	2.214
Ewe	XLM-R / AfroXLMR	3.400	2.883	2.606	2.522
Ewe	GhanaTok (BPE)	2.233	1.820	1.733	1.739
Ewe	GhanaTok-Unigram	1.733	1.452	1.328	1.334

7 Discussion

Answers to the research questions. General tokenizers split most Ghanaian words: 2.3 to 2.6 tokens per word and 64% to 72% of words split (RQ1). GhanaTok cuts fertility by 30.7% on Twi and 23.6% on Ewe against the best general tokenizer, with narrow intervals and a win on nearly every sentence (RQ2). One shared vocabulary of 8,000 tokens worked for both languages, with a small and significant gap between the two languages (RQ3).

Behaviour of the baselines. mBERT beats XLM-R and AfroXLMR on both languages. The reason was not tested; one plausible reason is that the XLM-R vocabulary, shared across about 100 languages, leaves little room for Ghanaian letters. AfroXLMR matches XLM-R exactly, consistent with a shared tokenizer vocabulary, so adapting a model to African languages did not alone fix the tokenization. ABENA, built for Twi, did not beat mBERT on fertility.

Why GhanaTok works, and where. The gains probably come from three

Table 8: Number of tokens used for example words.

Word	Lang.	BPE	Unigram	XLM-R	ABENA
Nkrumah	Twi	1	1	3	1
Mprempren	Twi	1	1	3	2
Ahwɛyie	Twi	3	3	5	5
Esiane	Twi	1	1	2	2
Bɔɔl	Twi	3	1	4	4
Afetɔ	Ewe	1	1	4	n/a
Dɔwɔnyawo	Ewe	2	1	6	n/a
Akantɔwo	Ewe	2	1	4	n/a
Asante	Ewe	1	1	2	n/a

things: letters kept inside words, reserved slots for common word parts, and a vocabulary spent fully on two languages. No test removed each part in turn, so the contribution of each part is unknown. The error analysis adds a trade-off: specialisation costs coverage. GhanaTok loses to mBERT mainly on English and mixed-language sentences. Real Ghanaian text often mixes languages, so a deployed tokenizer would need English coverage, for example by adding general English pieces or by combining GhanaTok with an existing vocabulary.

Practical meaning. A lower token count means lower cost and more usable context, if a model uses the tokenizer. For the same Twi text, general tokenizers would produce 44% to 48% more tokens than GhanaTok (BPE). Lower fertility is not the same as better understanding, and the sentiment probe does not show an accuracy gain. The evidence shows that GhanaTok removes a clear inefficiency, while the effect on model quality remains to be tested.

Claims not supported. The study does not show better downstream accuracy, better translation, or better handling of other domains or dialects. The cost figures are estimates. The vocabulary sweep supports but does not independently select the 8,000-token size.

8 Limitations

- **Small, narrow data.** The test sets hold 608 Twi and 347 Ewe sentences and the calibration sample 3,000 per language. The text leans toward dialogue, Bible text and civic notices, so results may differ on news, books or social media.
- **Not a fully fair contest.** GhanaTok was trained on the same corpus as the test sentences. The baselines were not adapted to the corpus, so part of the gain may be a domain advantage.
- **No ablations.** The pre-tokenizer and seed words were not removed one at a time to measure the separate effect of each.
- **Vocabulary sweep.** The sweep was run on the sealed test sets after 8,000

tokens was chosen. Sizes near XLM-R's 250,000 tokens were not tested.

- **ABENA and dialect.** ABENA covers Twi only. The Twi corpus has no dialect labels, and the Akuapem check uses a short keyword list.
- **One run.** One tokenizer was trained per setting, so variation from training is not reported.
- **Error analysis scope.** Sentences were inspected for Twi only, by counts and surface features. Native speakers did not rate whether splits are linguistically sensible.
- **Downstream scope.** The sentiment probe uses a linear model on short tweets, and the near-identical mBERT and ABENA results were not explained. Training a language model was outside this study.
- **Speed and price.** Timings come from one computer and different libraries. Cost figures assume one reference price and that a model uses the tokenizer.

9 Conclusion

Standard multilingual tokenizers split 64% to 72% of Twi and Ewe words and use 2.3 to 2.6 tokens per word. A small shared tokenizer trained on 7,621 sentences brings this to 1.656 for Twi and 1.750 for Ewe, and a Unigram version reaches 1.251 and 1.349. GhanaTok uses fewer tokens than mBERT on 98.4% of Twi test sentences, and the failures cluster in English phrases, loanwords and quotation marks. The most useful next steps are to add English and mixed-language coverage, to pretrain a small language model with GhanaTok and measure perplexity, training cost and task accuracy, to test other domains and dialects, and to ask Twi and Ewe speakers to review the splits.

Code and data availability. The repository `https://github.com/exornamb/GhanaToken` (final tag `v1.0-final`) holds the source code in `src`, nine numbered notebooks, saved metrics and figures in `reports`, the README with reproduction steps, the data card, model card, literature matrix, ethics statement, AI-use disclosure, failure log and reproducibility checklist. Text data are public and downloaded as described in Section 3.

AI-use statement. Claude (Anthropic) assisted with planning, code, wording, reference checking and document preparation. All results come from code run by the author. Numbers in this paper were matched to the saved CSV files, and the 2024 to 2026 references and ABENA were checked against arXiv or ACL Anthology pages. The full disclosure is in `ai_use_disclosure.md`.

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A Additional Results

Table 9: All paired tests, including the Unigram comparisons. “Base” is the baseline fertility and “GhanaTok” is the fertility of the first tokenizer listed. Cut is the percent reduction.

Lang.	Tokenizer	Baseline	Base	GhanaTok	Cut % [95% CI]	Holm p
Twi	GhanaTok (BPE)	mBERT	2.389	1.656	30.7 [29.85, 31.44]	$< 10^{-50}$
Twi	GhanaTok (BPE)	XLM-R / AfroXLMR	2.444	1.656	32.2 [31.45, 32.98]	$< 10^{-50}$
Twi	GhanaTok (BPE)	ABENA	2.419	1.656	31.5 [30.72, 32.30]	$< 10^{-50}$
Ewe	GhanaTok (BPE)	mBERT	2.291	1.750	23.6 [22.08, 24.95]	$< 10^{-50}$
Ewe	GhanaTok (BPE)	XLM-R / AfroXLMR	2.608	1.750	32.9 [31.61, 34.11]	$< 10^{-50}$
Twi	GhanaTok-Unigram	mBERT	2.389	1.251	47.6 [46.81, 48.43]	$< 10^{-50}$
Twi	GhanaTok-Unigram	XLM-R / AfroXLMR	2.444	1.251	48.8 [48.02, 49.62]	$< 10^{-50}$
Twi	GhanaTok-Unigram	ABENA	2.419	1.251	48.3 [47.48, 49.08]	$< 10^{-50}$
Twi	GhanaTok-Unigram	GhanaTok	1.656	1.251	24.5 [23.71, 25.23]	$< 10^{-50}$
Ewe	GhanaTok-Unigram	mBERT	2.291	1.349	41.1 [39.67, 42.39]	$< 10^{-50}$
Ewe	GhanaTok-Unigram	XLM-R / AfroXLMR	2.608	1.349	48.3 [47.05, 49.33]	$< 10^{-50}$
Ewe	GhanaTok-Unigram	GhanaTok	1.750	1.349	22.9 [21.90, 23.88]	$< 10^{-50}$

Table 10: Calibration sample (3,000 sentences per language).

Lang.	Tokenizer	Fert.	Tok/char	% split
Twi	mBERT	2.371	0.469	70.0
Twi	XLM-R	2.438	0.483	70.1
Twi	AfroXLMR	2.438	0.483	70.1
Twi	ABENA	2.402	0.475	71.7
Ewe	mBERT	2.303	0.445	64.7
Ewe	XLM-R	2.625	0.506	71.4
Ewe	AfroXLMR	2.625	0.506	71.4

Table 11: Fertility by vocabulary size on the sealed test sets.

Lang.	Method	4k	8k	16k	32k
Twi	BPE	1.777	1.656	1.585	1.585
Twi	Unigram	1.354	1.251	1.212	1.212
Ewe	BPE	1.846	1.750	1.712	1.712
Ewe	Unigram	1.494	1.349	1.296	1.296

Table 12: Whole-test-set encoding time over repeated runs, one computer.

Lang.	Tokenizer	Mean s/run	SD	Sent./s
Twi	GhanaTok-Unigram	0.0231	0.0043	26,363
Twi	AfroXLMR	0.1808	0.0036	3,362
Twi	ABENA	0.1972	0.0060	3,083
Twi	GhanaTok (BPE)	0.2916	0.1478	2,085
Twi	XLM-R	0.3055	0.0513	1,990
Twi	mBERT	0.3131	0.0339	1,942
Ewe	GhanaTok-Unigram	0.0239	0.0148	14,536
Ewe	GhanaTok (BPE)	0.1037	0.0052	3,345
Ewe	mBERT	0.1270	0.0091	2,732
Ewe	AfroXLMR	0.1297	0.0028	2,676
Ewe	XLM-R	0.1319	0.0051	2,632

Table 13: AfriSenti Twi sentiment probe.

Tokenizer	Features	Acc. %	Macro F1	95% CI
mBERT	3,896	65.02	0.562	0.5282, 0.5952
XLM-R	4,483	64.07	0.545	0.5129, 0.5779
AfroXLMR	4,483	64.07	0.545	0.5120, 0.5775
ABENA	3,886	65.02	0.562	0.5285, 0.5934
GhanaTok (BPE)	1,899	62.91	0.536	0.5026, 0.5697
GhanaTok-Unigram	2,133	62.91	0.536	0.5030, 0.5676

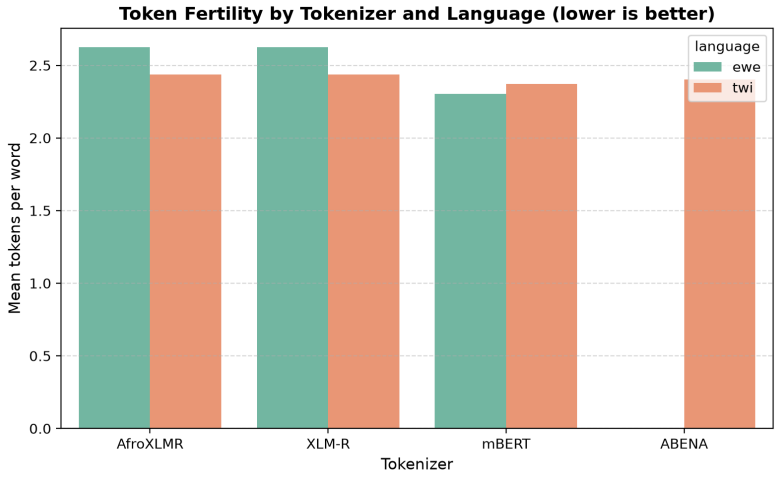


Fig. 4: Fertility of the existing tokenizers on the calibration sample.

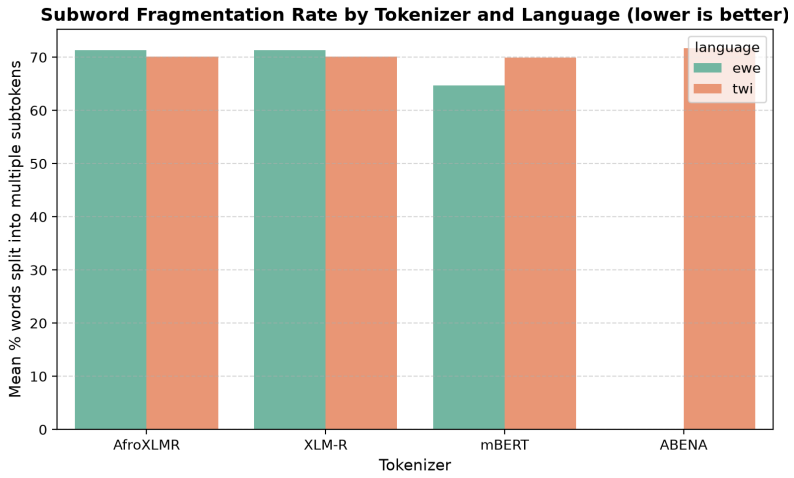


Fig. 5: Share of words split into several tokens on the calibration sample.

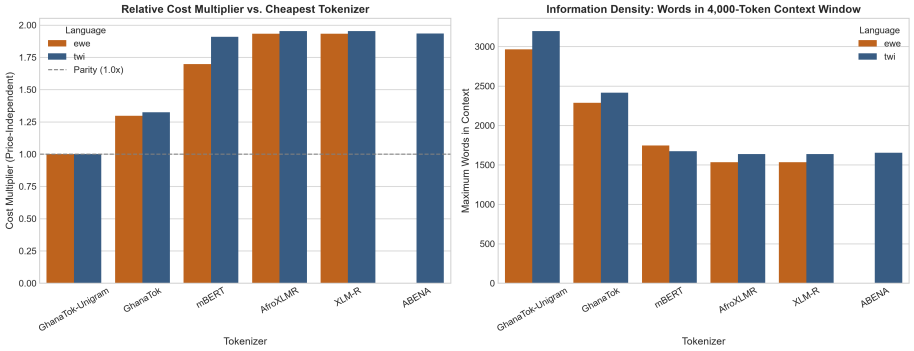


Fig. 6: Cost multiplier against the cheapest tokenizer, and words fitting in a 4,000-token window.